CALCULUS · UNIT 2: LIMITS AND CONTINUITY

Toll Road Pricing

Use limits and continuity to design fair prices for a new toll road and parking garage.

CP1

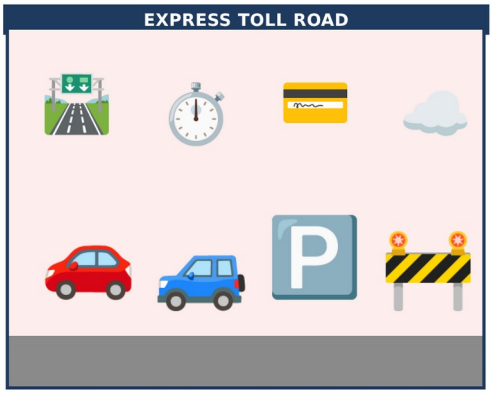
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

The state highway authority is opening a new express toll road with a parking garage at the end. Drivers complain when prices jump suddenly, and the authority wants prices that are fair and predictable.

You will use limits and continuity to analyze the current prices and propose better ones.



Garage

$3 per started hour

max $20 per day

Toll

Rush hour

price changes at 7 a.m.

Learning Goals

- I can evaluate limits graphically, numerically, and algebraically.
- I can determine continuity and fix discontinuities.
- I can use limits at infinity and the IVT in context.

Project Timeline

When	What you will do
Day 1	Discontinuities
Day 2	Continuous Tolls
Day 3	Pass vs. Pay
Final	Pricing Proposal + reflection

Part 1: Discontinuities

Day 1 · Use the Pricing Data Cards.

Pricing Data Cards

Garage	Toll Plan A	Toll Plan B
$3 for each hour	T(x) = 2 + 0.1x, x < 20	T(x) = 2 + 0.1x, x < 20
or part of an hour	T(x) = 5, x ≥ 20	T(x) = k + 0.05x, x ≥ 20

Monthly Pass	Traffic	Limit Table
$40 + $0.50 per trip	6 a.m.: 900 cars/h	f(x) = (x² − 9)/(x − 3)
	8 a.m.: 2,400 cars/h	

1. Classify the discontinuities of the garage price P(t) for 0 < t ≤ 6.

2. With the $20 daily maximum, write P(t) for 6 < t ≤ 24.

3. Find lim_{x→3} $\frac{x^2-9}{x-3}$ and classify the discontinuity of f at x = 3.

4. Redefine f to make it continuous at x = 3.

Part 2: Continuous Tolls

Day 2 · Toll Plan B.

5. Find k so that Plan B is continuous at x = 20. Show the three continuity conditions.

6. Design a toll $T(x) = ax^2 + b$ for $x \geq 20$ that is continuous at 20 and costs $8 at 40 miles.

7. Find lim_{x→20} of your design from both sides to confirm.

8. Is there a value of k that makes Plan A continuous? Explain.

Part 3: Pass vs. Pay

Day 3 · Monthly pass: \$40 + \$0.50 per trip. Average cost per trip A(n) = $\frac{40+0.5n}{n}$.

9. Find lim_{n→∞} A(n) and interpret.

10. Find lim_{n→0⁺} A(n) and interpret.

11. Use the IVT to show that the average cost equals \$3 for some number of trips between 10 and 30. Then find it.

Final Product: Pricing Proposal

Your final product must include:

- Classified discontinuities
- Continuous toll designs with justification
- Limits at infinity for the pass
- IVT argument
- Pricing proposal

Reflection

R1. What are the three conditions for continuity?

R2. How do limits at infinity help compare plans?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Limits and continuity conditions are applied correctly to analyze and design fair prices. Goes beyond the minimum with a creative, realistic choice.	Limits and continuity conditions are applied correctly to analyze and design fair prices.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Proposal includes graphs, limit work, continuity conditions, and a recommendation. Polished, organized, and visually engaging.	Proposal includes graphs, limit work, continuity conditions, and a recommendation. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.